

All-temperature exponential ergodicity for nonlinear spin exchange

A bounded-Gibbs-reweighting solution to arXiv:2511.05223, Section 4.5

Candidate full solution produced in run `fa_banach_001`

13 August 2026

Status. `candidate_full_solution_likely_valid_human_review_needed`. The argument removes the high-temperature hypothesis from the exact multi-component mean-field setting of Caputo–Morellini’s Theorem 1.7.

Source. P. Caputo and M. Morellini, *Kac’s Program and Relative Entropy Decay for Nonlinear Spin-Exchange Dynamics*, arXiv:2511.05223v1 (7 November 2025), Section 4.5, PDF page 34 [1].

1 Source question

The source asks whether exponential decay to equilibrium continues to hold for *every* symmetric interaction matrix J , allowing a rate that may deteriorate exponentially in the dimension. It proposes, as one route, a uniform-in-tilt covariance estimate for the associated canonical particle system. The full source passage is reproduced below.

4.5. Open problem: exponential ergodicity at all temperatures? We conclude with the discussion of some open questions. The exponential decay established in Theorem 1.7 holds under the assumption that the interaction matrix J is nonnegative definite and has spectral radius $\lambda(J) < 1/2$. The positivity assumption is, in principle, not restrictive, since one may freely modify the diagonal entries of J without affecting the corresponding Ising measure. The spectral radius condition, however, imposes that the interaction be sufficiently weak, which we interpret as a high-temperature assumption.

For instance, in the case of the Curie-Weiss model, where $J_{i,j} = \beta/n$ for all $i, j \in [n]$, the condition $\lambda(J) < 1/2$ imposes $\beta < 1/2$ which is a factor 2 off with respect to the critical value $\beta_c = 1$. It would be certainly desirable to have exponential decay with rate of order $1/n$ throughout the whole region $\beta \in (0, \beta_c)$ in this case. On the other hand, understanding the behavior of the system in the low temperature region $\beta > \beta_c$ remains an open problem. In particular, it is not even clear whether one should expect a slowdown or not for the nonlinear equation.

Nevertheless, by analogy with linear evolutions such as low-temperature Glauber or Kawasaki dynamics [29], one may ask whether exponential decay to equilibrium still holds for every interaction J , possibly with rates that deteriorate exponentially in n . Establishing such a result would amount to an exponential ergodicity property for the nonlinear equation, providing a quantitative counterpart to the plain convergence in Theorem 1.5.

A possible strategy toward this goal is to revisit Kac's program and show that a statement analogous to Theorem 1.10 remains valid for arbitrary interactions J , with some rate $\alpha \geq c(J)$ for a constant $c(J) > 0$ that may depend on J , possibly exponentially small in n . By repeating the reasoning in Section 3 and in particular, invoking Proposition 3.5, such a result would follow if one could prove that the tilted canonical Ising measure $\mathcal{T}_v \mu_{N,\rho}$ satisfies a covariance bound of the form

$$\lambda(\text{Cov}[\mathcal{T}_v \mu_{N,\rho}]) \leq C(J),$$

for some constant $C(J)$ that may grow with n but remains uniformly bounded in both the linear tilt v and the number of particles N . We conjecture that such a bound indeed holds for all interactions J . Using equivalence-of-ensembles estimates, as in Proposition 4.8, one can obtain such bounds for each fixed choice of the tilt v , uniformly in N . However, establishing estimates that are also uniform in the tilt parameter appears to be a rather challenging problem.

The proof below bypasses the proposed particle-system covariance conjecture. It transports the already-proved $J = 0$ nonlinear modified logarithmic Sobolev inequality (NMLSI) through a bounded Gibbs reweighting.

2 Definitions and statement

Let $\Omega = \{-1, +1\}^n$, let J be an arbitrary real symmetric $n \times n$ matrix, and set

$$E_J(\sigma) = \frac{1}{2} \sigma^\top J \sigma, \quad w(\sigma) = e^{E_J(\sigma)}, \quad \rho_J = \frac{\min_{\sigma \in \Omega} w(\sigma)}{\max_{\sigma \in \Omega} w(\sigma)} = e^{-\text{osc}_{\sigma \in \Omega} E_J(\sigma)}.$$

Fix a partition $\mathcal{A}$ of $[n]$ and the multi-component mean-field transport kernel $\mathcal{K}_{\mathcal{A}}$ of the source. Write

$$S_A(\sigma) = \sum_{i \in A} \sigma_i, \quad A \in \mathcal{A},$$

and let $\Gamma_{\mathcal{A}}$ be the space of fields constant on each $A \in \mathcal{A}$. The equilibrium family is

$$\mu_h^J(\sigma) = \frac{1}{Z_J(h)} e^{E_J(\sigma) + \langle h, \sigma \rangle}, \quad h \in \Gamma_{\mathcal{A}}.$$

For an $\mathcal{A}$ -regular law p , let $h_J(p)$ be the unique field for which p and $\mu_{h_J(p)}^J$ have the same $\mathcal{A}$ -magnetization profile, and put

$$I_J(p) = \text{KL}(p \parallel \mu_{h_J(p)}^J).$$

The source defines a nonlinear entropy production $\mathcal{D}_J(p)$ by writing $p = f\mu_{h_J(p)}^J$ in its H-theorem. We use the standard boundary convention for

$$\Phi(x, y) = (x - y) \log(x/y),$$

so $\Phi(x, x) = 0$ and $\Phi(x, 0) = +\infty$ for $x > 0$.

Theorem 1 (All-temperature exponential ergodicity). *For every real symmetric interaction matrix J , every partition $\mathcal{A}$ of $[n]$, and the multi-component mean-field kernel $\mathcal{K}_{\mathcal{A}}$, the nonlinear spin-exchange dynamics of [1] satisfies*

$$I_J(p_t) \leq e^{-\alpha_J t} I_J(p_0), \quad t \geq 0,$$

for every $\mathcal{A}$ -regular initial law p_0 , where the equilibrium entering I_J is fixed along the evolution by the conserved magnetization and

$$\boxed{\alpha_J = \frac{\rho_J^3}{4n} = \frac{1}{4n} \exp \left[-3 \operatorname{osc}_{\sigma \in \Omega} \left(\frac{1}{2} \sigma^\top J \sigma \right) \right] > 0.}$$

Thus the answer to the exponential-ergodicity question in Section 4.5 is affirmative. No definiteness or spectral-radius assumption on J is needed.

3 Proof intuition

Multiplication by the bounded weight $w = e^{E_J}$ maps the noninteracting exponential family exactly onto the interacting Ising family. This same change of density also has a particularly favorable effect on the Barker acceptance rule used by the collision kernel. For a proposed exchange, the two Gibbs pair weights appear through their harmonic mean. After the change of density, the nonlinear entropy-production integrand becomes the $J = 0$ integrand times a coefficient bounded below by ρ_J^2 . Separately, a bounded likelihood update can increase KL divergence by at most ρ_J^{-1} ; an accept/reject channel proves this in one line. The source's $J = 0$ NMLSI then transfers with total loss ρ_J^3 .

The information-projection step is important: the transformed law need not have the same magnetization as the naively transformed target. Instead, one projects the transformed law onto the whole $J = 0$ equilibrium family and maps that competitor back. This avoids any unjustified matching of fields.

4 Proof

For a probability q on Ω , define the bounded likelihood transform

$$T_w q(\sigma) = \frac{w(\sigma)q(\sigma)}{q[w]}, \quad q[w] = \sum_{\sigma} w(\sigma)q(\sigma).$$

This is a bijection of the probability simplex, preserves supports, and satisfies

$$T_w \mu_h^0 = \mu_h^J, \quad h \in \Gamma_{\mathcal{A}}. \tag{1}$$

4.1 KL comparison

Lemma 2 (Bounded likelihood updates). *If $a = \min w$ and $b = \max w$, then for all probabilities q, r on Ω ,*

$$\text{KL}(T_w q \| T_w r) \leq \frac{b}{a} \text{KL}(q \| r).$$

Consequently, if $p = T_w q$, then

$$I_0(q) \geq \rho_J I_J(p). \quad (2)$$

Proof. Apply to both q and r the same channel which, conditionally on $X = \sigma$, sets an auxiliary bit $B = 1$ with probability $w(\sigma)/b$. The accepted conditional laws are $T_w q$ and $T_w r$. The chain rule for KL gives

$$\text{KL}(q \| r) \geq \frac{q[w]}{b} \text{KL}(T_w q \| T_w r) \geq \frac{a}{b} \text{KL}(T_w q \| T_w r),$$

because the KL terms from the bit marginal and rejected branch are nonnegative. This proves the first assertion.

For the second, recall that the matching-magnetization member of either exponential family is its KL information projection. This follows directly by differentiating $\text{KL}(p \| \mu_h^J)$ in h ; its gradient is the difference between the sufficient-statistic means. Let $r = \mu_g^0$ minimize $\text{KL}(q \| \mu_g^0)$ over $g \in \Gamma_{\mathcal{A}}$. By (1), $T_w r = \mu_g^J$ is an admissible competitor in the interacting family. Hence

$$I_J(p) \leq \text{KL}(T_w q \| T_w r) \leq \rho_J^{-1} \text{KL}(q \| r) = \rho_J^{-1} I_0(q),$$

which is (2). □

4.2 Exact entropy-production comparison

Fix one ordered proposed exchange

$$x = (\sigma, \sigma') \mapsto y = (\tau, \tau').$$

The proposal is an involution and conserves the vector $S(\sigma) + S(\sigma')$. Set

$$A = w(\sigma)w(\sigma'), \quad B = w(\tau)w(\tau'), \quad Z = q[w].$$

All external-field factors therefore cancel between the two pair weights. The Barker acceptance probability from x to y is exactly $B/(A + B)$.

Lemma 3 (Collision comparison). *If $p = T_w q$, then*

$$\mathcal{D}_J(p) \geq \rho_J^2 \mathcal{D}_0(q). \quad (3)$$

Proof. Let $\mu = \mu_h^J$ for any $h \in \Gamma_{\mathcal{A}}$ and put $f = p/\mu$. In the source's entropy-production formula, the contribution of the accepted proposal $x \rightarrow y$, before the common proposal rate and common factor $1/4$ are applied, is

$$\mu(\sigma)\mu(\sigma') \frac{B}{A+B} \Phi(f(\sigma)f(\sigma'), f(\tau)f(\tau')).$$

Using pair-statistic conservation, the ratio of the two equilibrium pair weights is A/B . Substituting $p(\eta) = w(\eta)q(\eta)/Z$ and simplifying gives exactly

$$\frac{AB}{Z^2(A+B)} \Phi(q(\sigma)q(\sigma'), q(\tau)q(\tau')). \quad (4)$$

The rejected proposal contributes zero. For $J = 0$, the acceptance probability is $1/2$, so the corresponding coefficient of Φ is $1/2$. The ratio of (4)’s coefficient to $1/2$ is

$$\frac{2AB}{Z^2(A+B)} \geq \frac{\min(A,B)}{Z^2} \geq \frac{a^2}{b^2} = \rho_J^2.$$

The first inequality is the elementary harmonic-mean bound. Multiplying by the identical proposal rates and the identical factor $1/4$, then summing over all ordered input pairs and site proposals, proves (3). The calculation remains valid on the boundary of the simplex by the lower-semicontinuous convention for Φ , or by positive regularization. $\square$

4.3 Completion

The source proves Theorem 1.7 and its equivalent NMLSI for every partition whenever the interaction is nonnegative definite with spectral radius below $1/2$ [1]. At $J = 0$ its stated constant is exactly $1/(4n)$. Therefore, for every $\mathcal{A}$ -regular q ,

$$\mathcal{D}_0(q) \geq \frac{1}{4n} I_0(q). \quad (5)$$

If p is $\mathcal{A}$ -regular then $q = T_w^{-1}p$ is also $\mathcal{A}$ -regular, because the positive transform preserves its support. Combining Lemmas 2–3 with (5) yields

$$\mathcal{D}_J(p) g e \rho_J^2 \mathcal{D}_0(q) \geq \frac{\rho_J^2}{4n} I_0(q) \geq \frac{\rho_J^3}{4n} I_J(p).$$

This is the all-temperature NMLSI. The source H-theorem says that along the nonlinear evolution,

$$\frac{d}{dt} I_J(p_t) = -\mathcal{D}_J(p_t),$$

while the matching field is fixed by conservation of the $\mathcal{A}$ -magnetization. Grönwall’s inequality proves Theorem 1. $\square$

5 Verification and limitations

- The reusable script `code/verify_reweighting.py` exhaustively sums every ordered collision and site proposal on small Boolean cubes. On 400 randomized positive-law cases ($n = 2, 3$), it matched (4) to relative error at most 1.30×10^{-15} ; every dissipation, KL, information-projection, and final comparison had positive numerical slack.
- The theorem answers the all-temperature exponential-ergodicity question in the exact quantitative setting of Theorem 1.7: arbitrary partitions and their multi-component mean-field kernels. It does not claim the result for an arbitrary symmetric transport kernel.
- The source’s separate uniform covariance conjecture for the Kac particle system is not proved. The result shows that this conjecture is unnecessary for the nonlinear exponential-decay conclusion.
- The rate is deliberately crude. In particular, it does not recover the expected order- $1/n$ rate throughout the subcritical Curie–Weiss regime. It does establish the source’s requested positive $c(J)$, with exponential deterioration in n allowed.
- Diagonal modifications of J add a constant to E_J on the Boolean cube, so $\text{osc } E_J$ and the stated rate are invariant under them.

6 Novelty check and review recommendation

A bounded search was performed on 13 August 2026 through the run registry, solution, attempt, proof-gap, and archive indexes, followed by arXiv/web searches using arXiv:2511.05223, the exact title, both authors, the phrases “all temperatures”, “exponential ergodicity”, “uniform in the tilt”, and “tilted canonical Ising measure”, plus close spectral-independence and conditional-product terms. The source remains at v1 and no later paper or preprint explicitly resolving this question was found. Novelty is therefore plausible but not certified.

Human review is strongly recommended. The highest-value checks are: (i) the source normalization yielding coefficient $1/2$ at $J = 0$; (ii) the proposal-by-proposal simplification in (4); and (iii) the information-projection direction in Lemma 2. The included exact finite-state checker targets (i)–(ii).

References

- [1] P. Caputo and M. Morellini, *Kac’s Program and Relative Entropy Decay for Nonlinear Spin-Exchange Dynamics*, arXiv:2511.05223v1, 2025.